

beautifully Cucumber has taken care of every minute detail while designing this framework.

- Now copy the following code in the *step definition* file *TestCase.java*:

```
package stepDefinition;

import java.util.concurrent.TimeUnit;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import cucumber.api.java.en.Given;
import cucumber.api.java.en.Then;
import cucumber.api.java.en.When;

public class TestCase {

    WebDriver driver = null;

    @Given("^A\\s*([\\s\\S]*)\\s*" browser is opened$")
    public void browser_is_opened(String arg1) throws
    Throwable {
        if(arg1.equals("chrome")) {
            System.setProperty("webdriver.chrome.driver",
            "D:\\Selenium\\chromedriver.exe"); // file path of driver
            where it is stored.
            driver = new ChromeDriver();
            driver.manage().window().maximize();
            driver.manage().timeouts().
            implicitlyWait(30, TimeUnit.SECONDS);
        }
        @When("^A\\s*([\\s\\S]*)\\s*" is opened$")
        public void is_opened(String arg1) throws Throwable {
            driver.get(arg1);
        }
        @When("^A\\s*([\\s\\S]*)\\s*" is searched in search box$")
        public void is_searched_in_search_box(String arg1)
        throws Throwable {
            driver.findElement(By.id("lst-ib")).sendKeys(arg1);
            driver.findElement(By.xpath("//*[@id='tsf']/div[2]/
            div[3]/center/input[1]")).click();
        }

        @Then("^the first link in search results should be
        opened$")
        public void the_first_link_in_search_results_should_be_
        opened() throws Throwable {
            driver.findElement(By.xpath("//*[@id='rso']/div[3]/
            div/div[1]/div/div/h3/a")).click();
        }

        @Then("^browser is closed$")
```

```
public void browser_is_closed() throws Throwable {
    driver.close();
}
```

- Since we have defined the code, run the *runner* file and the corresponding steps will be performed according to the defined test case.
- Once execution is complete, you can see the execution status of the steps and scenarios in the console as shown in Figure 5.

```
Starting ChromeDriver 2.31.488763
Only local connections are allowed.
Aug 08, 2017 8:21:48 PM org.openqa.
INFO: Detected dialect: OSS

1 Scenarios (0[32m passed0[0m)
5 Steps (0[32m passed0[0m)
0m13.935s
```

Figure 5: Output of the script depicting one scenario and five steps being passed

So you can see how easily Cucumber can be configured with Selenium Web driver to implement the BDD framework. Using this framework, you can start your DevOps journey in the testing field, within your organisation.

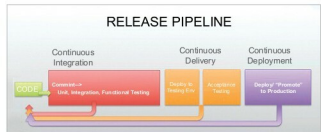


Figure 6: Complete release pipeline with CI/CD tools implemented (Image courtesy: Pinterest)

CI/CD tools and DevOps

This framework can also be integrated smoothly with continuous integration and continuous delivery (CI/CD) tools like Jenkins, TeamCity, Bamboo, etc, so that automated tests can be run every time developers check their code into a central repository; reports can then be published to the required stakeholders as and when required.

So we have discussed the shift from the Waterfall model to the Agile model as well as the simultaneous implementation of DevOps and the Agile methodology. Try this BDD inspired framework using Selenium to leverage your team's productivity and efficiency. 

By: Vinayak Vaid

The author works as an automation engineer at Infosys Limited, Pune. He has worked on different testing technologies and automation tools like QTP, Selenium and Coded UI. He can be contacted at vinayakvaid91@gmail.com.